

Peter Van Katwyk

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Professional Summary

PhD candidate in Scientific Machine Learning with 5+ years of experience building and deploying scalable ML systems across NASA, Meta, and academia. Strong background in deep learning, cloud-native ML (AWS, PyTorch), uncertainty quantification, and temporal and geospatial modeling. Skilled at translating complex research into production-ready tools that drive real-world impact.

Technical Skills

Programming: Python, SQL, Bash, MATLAB, R, Julia, C++

ML/AI: PyTorch, TensorFlow, Bayesian modeling, GenAI, Spatial/Temporal Modeling

Data & Tools: AWS (SageMaker, Lambda, API Gateway, DynamoDB, etc.), Docker, PostgreSQL, Git/Github

DevOps/MLOps: WandB, Flask, DVC, MLFlow, Containerization, Cloud Deployment, CI/CD

Experience

Research Scientist (PhD), NSF GRFP Fellow, Brown University – Providence, RI Sep 2021 - Present

- Built Bayesian deep learning emulators for geospatial ice sheet modeling that cut sea level projection error by **80%** and reduced inference time by **10x**, enabling faster, more reliable decision-making tools for climate planning
- Developed and maintained **ise**, a scalable production-grade open-source Python library for end-to-end data preprocessing, training, and evaluation of scientific ML models, integrated with PyTorch and HPC/GPU capabilities
- Collaborated across scientific and ML disciplines, and advanced methods for uncertainty quantification and published four first-author papers (*TMLR*, *GMD*, *JAMES*, *Nature Comm.*) with co-authorships in *Nature* and *The Cryosphere*

Machine Learning Engineer (Contract), Meta – Remote Aug 2020 – May 2025

- Led design, prototyping, and deployment of cloud-scale infrastructure for a geospatial ML platform supporting large-scale fiber network expansion
- Built and productionized **4** time-series and computer vision models and LLM-driven geospatial models for automated soil classification, permitting, and reporting—used by **150+** engineering teams at Meta and Vermeer
- Conducted training sessions on Data Science and ML best practices and deployment, improving adoption and maintainability across Meta and Vermeer teams

Data Scientist Intern, NASA – Mountain View, CA Summer 2024

- Conceived and developed a Reinforcement Learning framework, **aiatc**, in PyTorch for geospatial AI-based air traffic control, including collision avoidance, dynamic weather rerouting, and sector-aware path planning
- Delivered **6%** reduction in flight times during simulated weather disruptions; framework maintained over **99.7%** routing success rate in simulation tests, demonstrating robustness under real-world conditions
- Presented work at NASA-wide conference and earned *Outstanding Intern Presentation* award

Education

Brown University, PhD in Scientific Machine Learning Sep 2021 – May 2026

- **Advisor:** Karianne Bergen, **Dissertation:** AI for Climate, Bayesian NNs for Projecting Future Sea Level
- National Science Foundation GRFP Fellow

Brigham Young University, BS in Geological Sciences, Minor in Applied Statistics Sep 2017 – May 2021

Selected Publications

- **Van Katwyk & Bergen** (2025). HybridFlow: Quantification of Aleatoric and Epistemic Uncertainty with a Single Hybrid Model. *TMLR*. <https://doi.org/10.48550/arXiv.2510.05054>.
- **Van Katwyk et al.** (2025). ISEFlow v1.0: A Flow-Based Neural Network Emulator for Improved Sea Level Projections and Uncertainty Quantification. *Geoscientific Model Development*. <https://doi.org/10.5194/egusphere-2025-870>.
- **Van Katwyk et al.** (2023). A Variational LSTM Emulator of Sea Level Contribution from the Antarctic Ice Sheet. *JAMES*. <https://doi.org/10.1029/2023MS003899>.